Exercise Sheet 11

Discrete Mathematics by Hongfei Fu, 2023.11.20

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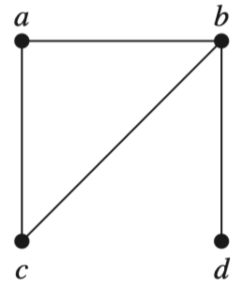
1. ([R], Page 650, Exercise 3,5,7,9) For Exercises 3–9, determine whether the graph shown has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops.

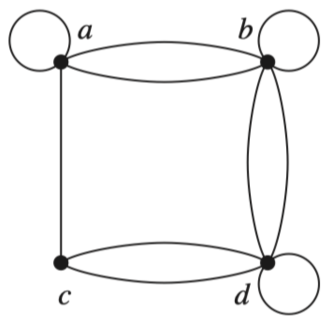
   3.

   Answer:
   The graph has:

   - Undirected edges.
   - No multiple edges.
   - No loops.

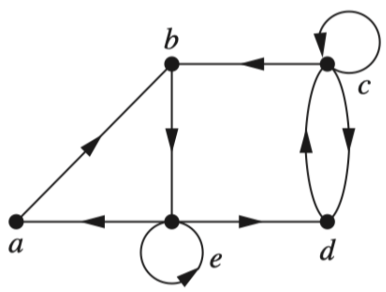
   5.

   Answer:
   The graph has:

   - Undirected edges.
   - Multiple edges.
   - Three edges.

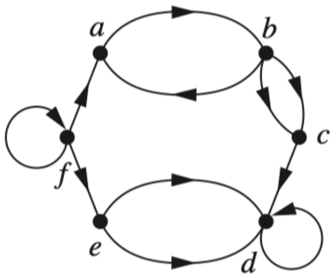
   7.

Answer:
The graph has:

- Directed edges.
- Multiple edges.
- Two loops.

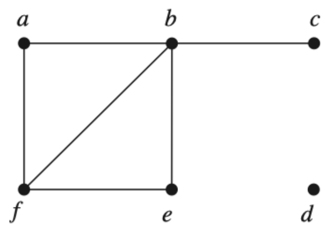
9.



Answer:
The graph has:

- Directed edges.
- Multiple edges.
- Two loops.

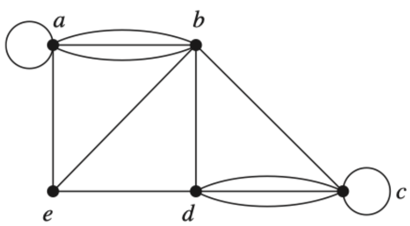
2. ([R], Page 665, Exercise 1, 2) In Exercises 1-2 find the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices.
1.



Answer:

- deg(a) = 2, deg(b) = 4, deg(c) = 1, deg(d) = 0, deg(e) = 2, deg(f) = 3
- isolated vertex: d
- pendant vertex: c

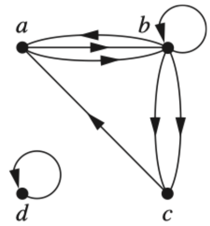
2.



Answer:

- deg(a) = 6, deg(b) = 6, deg(c) = 6, deg(d) = 5, deg(e) = 3
- There is no isolated vertex or pendant vertex.

3. ([R], Page 665, Exercise 8) In Exercises 8 find the in-degree and out-degree of each vertex for the given directed multigraph.



Answer:
$deg^+(a) = 2, deg^+(b) = 4, deg^+(c) = 1, deg^+(d) = 1$
$deg^-(a) = 2, deg^-(b) = 3, deg^-(c) = 2, deg^-(d) = 1$